

Program-1: Build a Docker Container from a Custom Dockerfile

Step 1: Create Project Directory

```
mkdir Program-1  
cd Program-1
```

Step 2: Create Dockerfile

Create a file named **Dockerfile** (no extension):

```
nano Dockerfile
```

Add the following content:

```
# Use a slim Python base image  
FROM python:3.9-slim  
  
# Set working directory  
WORKDIR /app  
  
# Copy requirements file and install dependencies  
COPY requirements.txt .  
RUN pip install --no-cache-dir -r requirements.txt  
  
# Copy application code  
COPY . .  
  
# Set Flask environment variables  
ENV FLASK_APP=app.py  
ENV FLASK_RUN_HOST=0.0.0.0  
ENV FLASK_RUN_PORT=5000  
  
# Expose Flask port  
EXPOSE 5000
```

Run the Flask application
CMD ["flask", "run"]

Save and exit (CTRL+O, ENTER, CTRL+X).

A screenshot of a terminal window showing a Dockerfile being edited in the nano text editor. The terminal title bar indicates the user is 1RV24MC109_syeda_nimra on a syedanimra-IdeaPad-Gaming-3-15IMH05 laptop. The Dockerfile content is as follows:

```
GNU nano 7.2 Dockerfile
FROM python:3.9-slim

WORKDIR /app

COPY requirements.txt .
RUN pip install --no-cache-dir -r requirements.txt

COPY . .

ENV FLASK_APP=app.py
ENV FLASK_RUN_HOST=0.0.0.0
ENV FLASK_RUN_PORT=5000

EXPOSE 5000

CMD ["flask", "run"]
```

Step 3: Create Python Application File

Create **app.py**:

```
nano app.py
```

Add the following code:

```
from flask import Flask
```

```
app = Flask(__name__)
```

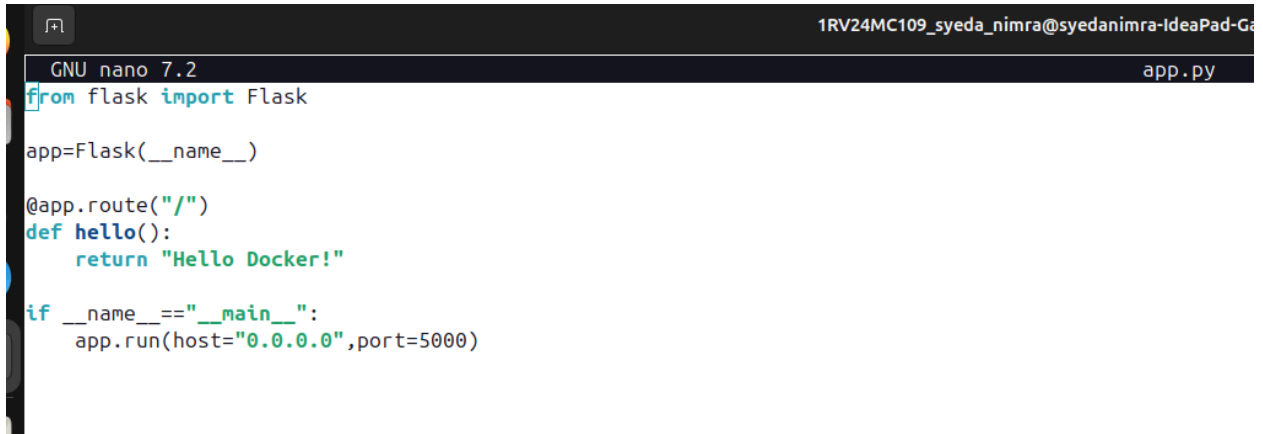
```
@app.route("/")
```

```
def hello():
```

```
    return "Hello, Docker!"
```

```
if __name__ == "__main__":  
    app.run(host="0.0.0.0", port=5000)
```

Save and exit.



```
GNU nano 7.2 app.py  
from flask import Flask  
  
app=Flask(__name__)  
  
@app.route("/")  
def hello():  
    return "Hello Docker!"  
  
if __name__=="__main__":  
    app.run(host="0.0.0.0",port=5000)
```

Step 4: Create requirements.txt

nano requirements.txt

Add:

flask

Save and exit.

Step 5: Build Docker Image

Run the Docker build command:

```
docker build -t program-1 .
```

```

1RV24MC109_syeda_nimra@syedaninra-IdeaPad-Gaming-3-15IMH05:~/program1$ nano Dockerfile
1RV24MC109_syeda_nimra@syedaninra-IdeaPad-Gaming-3-15IMH05:~/program1$ sudo docker build -t program-1 .
[+] Building 1.6s (10/10) FINISHED
=> [internal] load build definition from Dockerfile
=> => transferring dockerfile: 267B
=> [internal] load metadata for docker.io/library/python:3.9-slim
=> [internal] load .dockerignore
=> => transferring context: 2B
=> [1/5] FROM docker.io/library/python:3.9-slim@sha256:2d97f6910b16bd338d3060f261f53f144965f755599aab1acda1e13cf1731b1b
=> [internal] load build context
=> => transferring context: 328B
=> CACHED [2/5] WORKDIR /app
=> CACHED [3/5] COPY requirements.txt .
=> CACHED [4/5] RUN pip install --no-cache-dir -r requirements.txt
=> [5/5] COPY . .
=> exporting to image
=> => exporting layers
=> => writing image sha256:c1eef2f9232cf7308c2d8ed0d382cb56a6c7fb0b223f8a89ce7508d6b06aa89f
=> => naming to docker.io/library/program-1

```

Verify image creation:

docker images

Step 6: Run the Docker Container

docker run -d -p 5000:5000 --name flask-container program-1

```

1RV24MC109_syeda_nimra@syedaninra-IdeaPad-Gaming-3-15IMH05:~/program1$ sudo docker run -d -p 5000:5000 --name flask-container program-1
64cc1b2420b273e63f87b45a22167a5289e852bef6808c95a17d06bdee825025
1RV24MC109_syeda_nimra@syedaninra-IdeaPad-Gaming-3-15IMH05:~/program1$

```

Check running containers:

docker ps

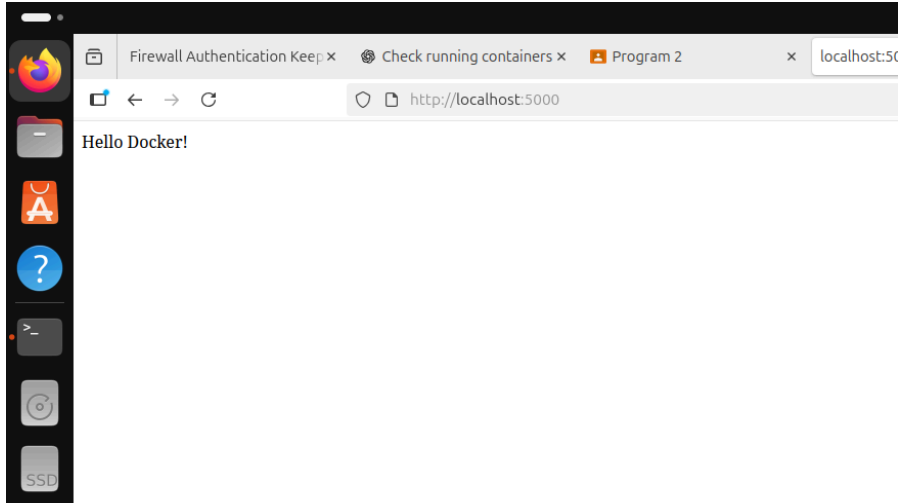
Step 7: Test the Application

Open a browser and go to:

<http://localhost:5000>

Output displayed:

Hello, Docker!



Step 8: Stop the Container

`docker container stop <container-id>`

Step 9: Remove the Container

`docker container rm <container-id>`

Step 10: Remove Docker Image

`docker image rm program-1`